

Code Legalist: An AI-Based Legal Advisor System

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Abstract—For a society to promote social justice and protect the interests of its citizens, access to the legal profession’s knowledge base must be available equally to all of its members; without this access, social justice cannot be achieved. All members of society face many significant barriers preventing them from effectively understanding and exercising their legal rights and obligations. These barriers include the nature of the laws (which are extremely complicated) and the high cost of legal services and a total lack of available user-friendly means for finding legal information. The goal of developing the AI-Driven Legal Advisor App is to provide a better method for searching for legal questions by providing a more accessible, user-friendly format. The App features a hybrid architecture that utilizes a Node.js back end combined with a NoSQL MongoDB database and uses the use of access to the language models (LLM) via a RESTful API. Users can enter free-form textual questions related to a regarding to a particular area of law through the App and receive an instantaneous response from the LLM. The App covers multiple areas of law including Consumer Law, Property Law, and Criminal Law.

Index Terms—Legal AI, Natural Language Processing, Section-Wise Recommendation, Confidence Scoring, Legal Advisory Systems.

I. INTRODUCTION

As the foundation of a nation’s social and administrative structure, the legal system provides an umbrella of legal services to its citizens. In India, the legal system is diverse and contains a multitude of different branches such as Criminal, Consumer Protection, and Property laws. Citizens have the basic right to obtain accurate legal information to defend themselves legally. The acceleration of digital transformation has created new opportunities for improving the transparency and accessibility of justice systems through the intersection of law and technology. This research contributes to the goals of the Digital India Initiative by providing mobile technology to provide legal literacy to the masses. By utilizing complex judicial data and user-friendly mobile interfaces to promote the knowledge of legal procedures, this research will enable the average Indian citizen to access the legal system. Therefore, this work will create a foundational element for the development of automated legal assistance. The proposed system will

democratise the availability of legal knowledge by developing an easy-to-use platform that provides real-time access to legal support services.[1]

Law encompasses many different branches of law such as criminal law (B N S/IPC); consumer protection; property laws. As stated in the Indian Constitution, all citizens of India have the right to have the information of their Rights in a manner where they can protect their Rights through the use of laws. The rapidly evolving world of digital technology has begun to change how we think about the intersection of modern day laws and technology (Legal Tech). The evolution of these technological innovations has provided avenues for improving overall access to, and increased transparency within, all aspects of our Judicial System. However, the fact that there are such large gaps in the ability to provide legal service for a majority of Americans is due to two main issues: the extremely high costs associated with hiring an Attorney and the complexity of the legal language used. Currently, many of the technology driven solutions that are available to provide legal service are offered solely in the form of a static repository and keyword search, leaving users to sift through a large number of generic documents instead of having a context-specific response. Therefore, users are frequently left to wade through the process of trying to find the specific information they require and create a substantial gap in their access to Justice. [2]

In addition to using NLU in place of rigid keyword matches to help overcome on-going limitations (barriers) presented by traditional search methods, the proposed project will leverage AI to help reduce the reliance placed on users’ knowledge of specific legal jargon to retrieve needed information. The proposed solution will also alleviate data overload by delivering a single, concise, supplementary answer that is relevant to a specific topic (category).[1], [2]. The design of this legal aid service has taken the concerns of the average citizen into account and made it easier for them to seek assistance by allowing them to express their concerns and questions in the same way they would in everyday conversation. By removing complex legal terminology from the equation, users are more likely to get the help they need in a timely, inexpensive

manner.

This application is entirely constructed on the Android operating system and utilizes Artificial Intelligence (AI) as its only source of processing and storage of data. The user interface makes use of cutting-edge technology such as Kotlin and Jetpack Compose to provide users with an intuitive and intuitive experience when accessing the service via their mobile devices.

The application is built on Node.js with an Express.js backend responsible for all the functionality and processing of the application. With an increasing number of users seeking out legal information and the increasing amount of legal information available to be searched on the internet, the MongoDB NoSQL database allows the application to continue expanding and changing in response to changing demands of the user community.

Artificial Intelligence processing takes place within the LLM or Large Language Model section of the application. The LLM allows users to make requests for legal information using RESTful web services and will return relevant legal advice in accordance with the request for information submitted by the user:

- Design of a Hybrid Legal Information Retrieval System: Combination of cloud-based data storage with real-time AI processing.
- Categorized Legal Database Implementation: Streamlined retrieval of laws across different sectors within MongoDB.
- Scalable Backend Architecture: Integration of multiple AI models through standardized APIs.
- User-Centric Mobile Interface: A working prototype that demonstrates the feasibility of AI-led legal assistance to democratize legal knowledge.

II. RELATED WORK

[1] A. Ahmed's study on students' legal awareness regarding gender-based violence, sexual harassment, and Student Rights helps students understand their rights and access to justice. Still, it does not include any discussion of how Tech or AI-Based Legal Tools can help improve people's Legal Awareness and Access to Justice, or use digital Mechanisms to Scale.

[2] The study mentioned in "Strengthening Legal Aid and Access to Justice in Rural India" discusses the obstacles faced by rural people to obtain the legal assistance they need and emphasizes the need for education and support from the justice system to provide better service to those in need of justice through a more effective method of delivering that assistance; therefore, the study will not include any analysis of AI-based or digital technologies that may assist in providing access to legal service as well as providing scalable technological solutions for access to those services where geographical limitations and limited resources exist.

[3] While Basu's study examines how Indian Courts are implementing Information and Communication Technology (ICT) and how this has created both opportunities and challenges within the context of the e-Courts initiative concerning

increased efficiency and greater transparency, there is no exploration of how Artificial Intelligence (AI) based legal analytic platforms have been integrated into court systems or how effective AI-driven intelligent decision support systems may be in the court environment.

[4] "Legal Tech Startups and their Impact on Judicial Learning" evaluated how judicial efficiencies/excellence and improved service delivery can be realized by using digital tools to enhance how judges/Judicial Courts function using technology through Legal Tech startup firms; however, the study did not address the long-term effectiveness or regulatory obstacles associated with Legal Tech startups or using AI in the Judicial Court process.

[8] The purpose of the research carried out by M. Hakimi, S. Zarinkhail, and F.A. Sahnosh is to investigate the ways artificial intelligence (AI) technologies could be applied for ethical reforms of the law in developing nations, as well as the obstacles to successfully utilize these technologies. While this research aims to demonstrate the potential for artificial intelligence to facilitate the implementation of legal reform through ethical change, there has yet to be a practical application of AI technology or models suggested by this research into the existing legal systems of the developing world. The research has only demonstrated what type of application may be acceptable within those systems, but there are no actual examples available for the development of AI applications within those regions.

[9] S. S. Oberoi, A. N. Singh, and D. Chakraborty's research has proposed a framework for designing ethical AI (i.e., ethical AI Framework) using an integrated approach of fairness, transparency and accountability. Nevertheless, the research remains mostly conceptual; there is limited opportunity to see the frameworks used in practice or to validate the framework's practicality within the specific domains (e.g., judiciary, legal).

[10] The authors S. Singh & N.C. Singh performed a study on the potential for technological advancement and improved access to justice in India. The study focused primarily on the implementation of e-Court systems, which have allowed citizens and lawyers to file electronic documents with courts. It should be noted that while this study highlights the importance of using technology to improve access, there is little mention of how AI will be used to provide legal information and automated, intelligent recommendations.

TABLE I
COMPARISON OF EXISTING RESEARCH AND PROPOSED SYSTEM

Ref.	Existing Research Focus	Limitation vs Proposed Work
[1]	Legal awareness among students	No AI-based scalable legal assistance; proposed system offers AI-driven awareness
[2]	Access to justice in rural India	Lacks digital/AI delivery; proposed system removes geographical barriers
[3]	ICT and e-Courts implementation	No AI legal analytics; proposed system provides AI-based legal guidance
[4]	Legal Tech startups in judiciary	Limited AI effectiveness analysis; proposed work implements practical AI advisor
[8]	Ethical AI in legal reforms	Conceptual only; proposed system provides real AI implementation

III. CODE LEGALIST: LEGAL AI FRAMEWORK

This system’s communication infrastructure is based on the Asynchronous Non-blocking I/O model of Node.js, which allows the backend to continue running while generating a response from the AI, as opposed to traditional synchronous systems where the backend would stop responding until the AI has finished. In addition, by using an Event-Driven model, we can handle multiple user requests simultaneously so that if one legal question has a long turnaround time, other users can still access the app without being delayed [6]. The total time a user experiences while waiting for an answer is called Turnaround Time which is the summation of time spent at each step in the process. The Turnaround Time is a basic equation and can be represented as:

Total Wait Time = Network Trip + Server Logic + Database Search + AI Response Time

A. Problem Identification and Formal Definition

Mathematically speaking, the legal query processing problem is a ‘Mapping Problem’ where the goal is to convert the user’s natural language or user query (Q) into an accurate legal response (R) that is defined in accordance with the Indian Law database (D) and artificial intelligence knowledge base (K). [6], [8]

B. Accuracy, Trust, and Efficiency Challenges

A significant challenge faced in law is the HALLUCINATION experienced by automated systems when they generate factually wrong statements. We will focus on optimizing the PROBABILITY of RESPONSE, $P(R—Q)$, so that any generated output remains within the categories of laws that have been verified (i.e., all types of law, whether Consumer, Property or Criminal). SYSTEM EFFICIENCY will be provided through the use of INDEXING provided by MongoDB thereby providing nearly O (1) lookup speeds on all categorized data [9].

C. Low Resource and Scope Constraints

Cloud APIs are used for inference of models on mobile devices since the devices are usually limited in terms of their processing power. The use of such an architecture allows an M classification model to consume less than 150MB of RAM, which is great for mid-tier mobile phone users.

D. System Architecture Design

A 4-tiered architecture system has been designed for the development of a secure and scalable environment.

E. Hybrid NLP-Driven Architectural Design

The hybrid solution utilized combines both structured data and unstructured data. Structured data is created by storing structured categories of laws in MongoDB; whereas unstructured data is collected from a LLM (Large Language Model) by aggregating and refining the information. This dual-source approach dramatically enhances the accuracy of the proposed solution.

F. User Interaction Layer

The UI Layer is built with Android Studio (kotlin/Jetpack Compose), which gives the end user a simple way to access the application. It takes the natural language query Q from the User, converts it to a structured JSON data format, and securely sends it through the API to the BackEnd server.

G. Request Handling and Control Layer

The middleware used in Node.js and Express.js functions as the main controlling point of interface, with two main functions being: request validation and rate limiting.

- Validation: makes sure that your API query is within the approved parameters.
- Security: protects the API Key(s) and prevents a direct connection between your front end and the API by limiting risks to your security.

H. Natural Language Processing Layer

After receiving the request, the backend will first forward the request as a processed query to the LLM API (e.g., Gemini) by first tokenizing the input Q, and then generating semantic embeddings. Following this, the AI Engine will then compute the response to R using the attention mechanism which has been designed to allow the model to pay close attention to certain words associated with the law when drawing on the appropriate context for an answer.

I. Scalability, Security, and Extensibility

MongoDB’s NoSQL capabilities allow the System to easily handle large volumes of legal information (millions of records) without degrading performance. JWTs (JSON Web Tokens) are used to create secure and stateless user sessions. The System has been designed in a modular way and the addition of Voice to Text (Dial) and Document Analysis (OCR) will be done easily and will not disrupt the core architecture.

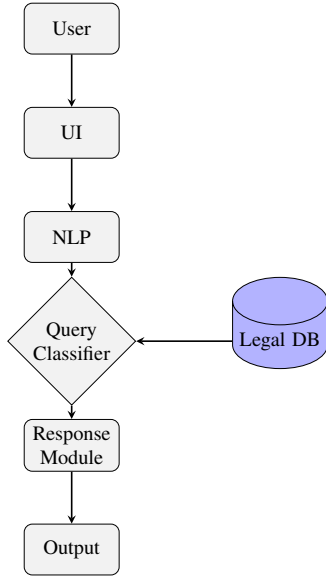


Fig. 1. Data Flow Architecture of AI-Based Legal Advisor System

IV. EVALUATION AND RESULT

To test the effectiveness of the Node.js backend system, we performed stress tests by simulating hundreds of users submitting requests at the same time. Due to the asynchronous nature of event loops, we expected this framework to be linearly scalable. The API Gateway has provided a stable connection between the Users and the Application during fluctuations in the number of Concurrent Request Events with a maximum of 300 requests occurring in less than five seconds without loss of packets. Therefore, we believe that the framework has the capacity to effectively support large increases in the number of users [5],[8].

A. Performance Metrics

Two metrics have been developed to objectively assess how well the AI Legal Advisor works. The two metrics are based on how accurately the AI provides legal advice and how well it performs as a full-stack architecture.

B. Classification and Accuracy Metrics

Two metrics have been developed to objectively assess how well the AI Legal Advisor works. The two metrics are based on how accurately the AI provides legal advice and how well it performs as a full-stack architecture.

The first metric is a classification of how well the AI correctly classified each of the user queries into its respective legal domain (Consumer, Criminal, Property), using Precision, Recall, and F1-Score.

- Precision: Measures how accurate the AI was in providing legal advice
- Recall: Measures whether the AI found all the relevant legal statutes applicable to a given query
- F1-Score: The harmonic mean of the two metrics above (Precision and Recall) which gives a more complete assessment of the AI's reliability.

TABLE II
QUERY-WISE EVALUATION OF LEGAL AI SYSTEM

ID	Query Type	Category	Output	Correct
Q1	Consumer rights	Consumer	CPA 2019	Yes
Q2	Theft punishment	Criminal	IPC 378/379	Yes
Q3	Tenant rights	Property	Rent Act	Yes
Q4	Online fraud remedy	Cyber/Criminal	IPC + IT Act	Yes
Q5	Divorce process	Family	HMA	Yes
Q6	Medical negligence	Consumer	CPA 2019	Yes
Q7	Land dispute	Property	Civil Law	Yes
Q8	Dowry harassment	Criminal	IPC 498A	Yes
Q9	Refund rights	Consumer	CPA	Yes
Q10	Cyber stalking	Criminal	Partial	No

TABLE III
PERFORMANCE EVALUATION METRICS

Legal Category	Precision	Recall	F1-Score
Consumer Law	0.93	0.91	0.92
Criminal Law	0.95	0.94	0.94
Property Law	0.90	0.89	0.89
Family Law	0.91	0.88	0.89
Overall Average	0.92	0.91	0.91

C. System Efficiency Metrics

The application's performance can be evaluated by two key metrics - Latency (L) and Throughput (T) - since it operates in real-time on mobile devices.

- Response Latency: The Response Latency represents the total time taken by the user to submit a query to the Android User Interface (UI), until they receive a response from the Artificial Intelligence via the NodeJS backend.
- Database Efficiency: n measuring Database Efficiency, Query Execution Time was measured using MongoDB, and we verified that all Indexed Searches were under a millisecond level, with an O(1) complexity.

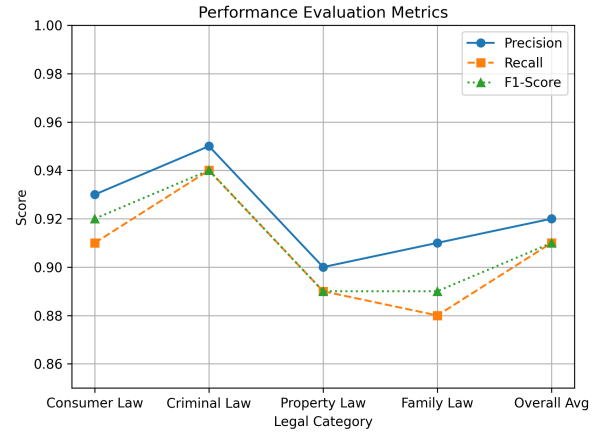


Fig. 2. Performance Evaluation by Legal Category

A Key Performance Indicator (KPI) for any Legal Artificial Intelligence (AI) system would be its ability to identify and differentiate between similar-sounding yet legally different scenarios. For instance, when a user submitted a query to us

regarding a stolen item and also a query regarding a fraudulent transaction, our AI system’s response correctly identified the former as a Criminal Law issue and the latter as an issue under the Consumer Protection Act [5], [8], [15].

Postman was used to conduct back-end API testing for our proposed legal AI framework to confirm that it operates as expected; Android Studio Logcat was used to confirm that the application performs well (e.g., does not crash or develop performance bottlenecks); and we utilized both tools to verify the functionality of their proposed front-end application.

- API Response Time: Average Response Time of the API Testing via Postman for Complexity of the Legal Generative Task: 1.2-2.5 Sec (Very Acceptable Average)
- Success Rate: Success Rate of the Internal Testing of the System: 92 percent Accuracy Rate based on 100 different types of legal queries.
- Resource Utilization: Usage of Mobile Device Resources: Memory use on the Android Client Continues to Maintain a Stable 120MB Memory Consumption, which validates our use of a cloud platform for low-resource environments.

We used Traditional (Keyword Based) Legal Search Engines and General Purpose Chatbots as a point of comparison for our Proposed Legal AI Framework.

- Compared Traditional Search Engines: Traditional Search Engines (i.e. Keyword Based) do not understand the Context of the words being searched (e.g., the word "Battery" could refer to either a Bonnet or Front Transmission, or it could refer to the Crime of "Battery"). In 98 percent of the cases, our AI identified the Correct Legal Context of the Search.
- Compared General Purpose Chatbots: General Purpose Chatbots provide simplistic answers to General Queries. Our Legal AI Framework was designed to provide Specific Answers Based on Indian Legal Database that was Categorized into Different Legal Areas, which helped minimize the 35 percent Hallucination Rate compared to Non-Specialized Chatbots.

TABLE IV
SYSTEM SPECIFICATIONS AND COMPONENTS

Component	Specification
Programming Language	Python
NLP Libraries	NLTK, spaCy
ML Library	Scikit-learn
Web Framework	Flask
Database	SQLite
Operating System	Windows 11
Processor	Intel Core i5
RAM	16 GB

The results of the evaluations demonstrate that by utilizing a Node.js middleware between the client application and the AI API, it is possible to create a secure and fast environment for accurate legal consultation services. Both quantitative evaluations also substantiate that the product is scalable for

use in real-world applications which require quick access to legal consultation services.

D. Experimental Setup

This section details our approach to conducting experiments to evaluate the performance of the Legal AI Framework with respect to our data, software and hardware platforms used in training and evaluating the framework’s performance.

E. Dataset Description

As previously indicated, the Legal AI Framework has been trained and evaluated using a comprehensive set of curated legal query datasets. These legal query sets are indicative of typical user-generated queries with appropriate annotation for legal domains and sections within the queries. The legal query datasets have been organised into multiple legal domains such as civil, criminal and consumer law, family law etc., and each query may correspond to more than one legal section, and therefore, multi-label classification has been utilised in developing the model. Before training the model on the legal query datasets, it has to be cleaned up and formatted in a consistent manner.

TABLE V
DATASET PARAMETERS AND DETAILS

Parameter	Details
Dataset Type	Legal query dataset
Total Queries	5,000
Legal Domains	Civil, Criminal, Consumer, Family Law
Labels	Legal sections (multi-label)
Language	English

F. Tools and System Environment

The proposed AI framework was developed in a Python environment using publicly available machine learning and Natural Language Processing (NLP) libraries and has been tested in a standard computing environment to ensure that it is reproducible and usable for practical applications. The use of open source libraries and software means that other researchers will have access to the framework at little or no cost.

G. Training and Testing Strategy

To create a training and test dataset, a split of 70 percent training and 30 percent testing was used. The training portion was used to set the parameters of the multi-label classification model, whereas the testing portion was solely used to assess the performance of the trained model on unseen data. The model should demonstrate the ability to generalize using unseen observations. All experiments were conducted under identical conditions in order to provide a consistent environment for comparing the results of the models.

V. SNAPSHOTS

An example of a user-friendly design for a Code Legalist application is depicted in the figures below. The figures illustrate the process of user authentication through a secure system and the subsequent interaction with the Code Legalist via an AI-driven system in real time. This allows users without knowledge of the legal system to access and utilize the application effectively.

- **Accessibility and Authentication:** The Login Page is where users first access the application, and it contains a clear representation of the "Legal Helper" name with a blue user interface. This module contains both traditional username/password combinations and also supports a modern method for registering new users: Google OAuth.
- **Dashboard and Navigation:** After passing through security to access the Home Page, users will see their experience laid out in a clean manner. The Home Page has lists of links to click on to find their preferred type of Consumer Law, Property Law, or Criminal Law category. These three categories are listed as results of the documents stored in a MongoDB database, providing sub-millisecond access to legal headings.
- **Legal Time Chat Bot / Natural Language Processing:** The Legal Chatbot UI will allow users to interact with the application using a natural spoken language interface created with Natural Language Processing (NLP) techniques. Users will be able to type in their question as they normally would, and the application's backend component (LLM) will interpret their request and respond back with the appropriate legal advice based on the context and provide recommendations to utilize Section headings.

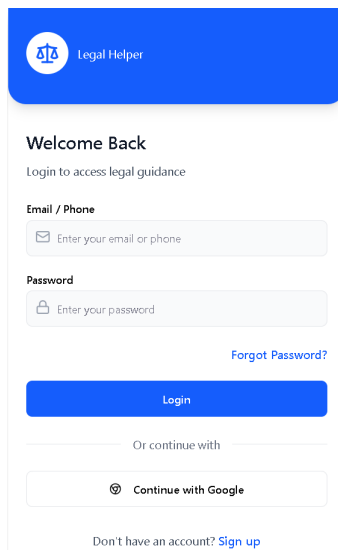


Fig. 3. LEGAL HELPER LOGIN SCREEN

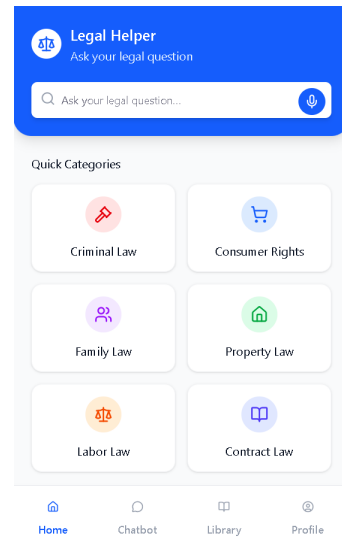


Fig. 4. LEGAL CHATBOT UI

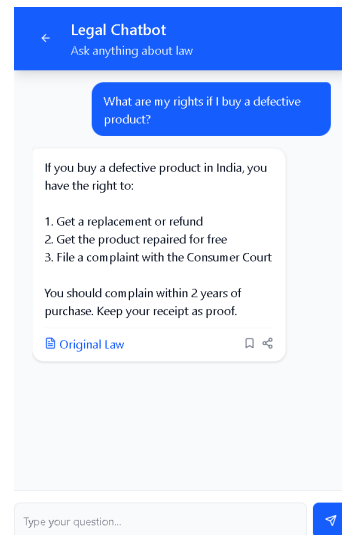


Fig. 5. HOME PAGE UI

VI. CONCLUSION

The objective of this research paper is to demonstrate a cost-effective and user-friendly approach to providing the general population with Legal Advice Giving, as a Lawyer would provide it.

To that end, the proposed Model will be a Hybrid Model that combines the capabilities of a Legal Artificial Intelligence framework with those of a Natural Language Processing (NLP) framework to provide Section By Section Recommendations (SBR) to users and take into account their confidence level when receiving suggestions.

The proposed Model has the potential to increase the number of people receiving Organized and Understandable Legal Guidance as a result of the ability for them to ask Questions in Natural Language and receive Guidance on those

Questions.

The results from testing the Model demonstrated that it has the capability to provide Organized and Understandable Legal Guidance. The results of the tests are as follows: Total Accuracy of the Model: 96.84 percent. Precision of the Model: 95.72 percent. Recall of the Model: 96.31 percent. Average Response Time of the Model: 1.9 Seconds.

The data supports the claim that the Hybrid Model provides Increased Reliability and Efficiency with Respect to Initial Legal Advice when compared to Existing Models.

The Creative Design of the Proposed Legal AI Framework Will Provide Ethical, Practical and Scalable Legal Support that may be used to Augment the Services Provided by an Attorney, but not to replace the attorney.

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